parameters as encoder-decoder models. In this paper, we make the simple choice to use a pretrained encoder-decoder model, here T5 (Raffel et al., 2020), entirely discard its decoder and finetune its encoder for regression and classification tasks. This has many desiderata: it allows us to cut the network size in half and importantly enables us to finetune on longer sequences and therefore account for longer-term dependencies in the crystal descriptions. With this simple yet effective choice, our approach outperforms not only large encoder-only models such as MatBERT, but also state-of-the-art GNN-based crystal property predictors such as ALIGNN.

Contributions. We make two main contributions, as summarized below:

- We collect, curate, and make public a benchmark dataset that contains approximately 144K crystal text descriptions and their properties.
- We propose LLM-Prop, an efficiently finetuned network that allows us to achieve state-of-the-art performance in crystal property prediction, outperforming the current best GNN-based crystal property predictors.

2 Related Work

GNNs for crystal property prediction. One of the main GNN-based crystal property predictor is CGCNN (Xie and Grossman, 2018). CGCNN uses a convolutional neural network (CNN) on top of node embeddings from a crystal graph to learn the interactions between atoms in the crystal and predict crystal properties. Although CGCNN outperformed classical methods (Jain et al., 2011; De Jong et al., 2015; Kirklin et al., 2015), it doesn't incorporate the many symmetries that crystals abide by. Several follow-up works aimed to incorporate those symmetries. Chen et al. (2019) proposed MEGNet, an architecture that incorporates crystal periodicity and only relies on GNNs to represent crystals. MEGNet outperforms CGCNN on various tasks; however, it still fails to account for critical information such as bond angles. ALIGNN (Choudhary and DeCost, 2021) was proposed to explicitly incorporate bond angles in addition to the other information accounted for by MEGNet and achieves state-of-the-art results compared to previous GNN approaches.

Finetuned LLMs for crystal representation. There have been very recent efforts using LLMs to learn representations of crystalline materials from their text descriptions. Qu et al. (2023) proposed an NLP-based crystal discovery framework that uses text embeddings as crystal representations. Crystal text descriptions were first fed into the in-domain pretrained LLMs to get embeddings. Then these embeddings were used to finetune a regression model for crystal property prediction. Finally, candidate crystals are recalled and ranked on the basis of the targeted properties. Several aforementioned in-domain pretrained LLMs were explored and MatBERT gave the best performance, a similar finding with Song et al. (2023). Even though these works rely on text to learn crystal representations, they mainly focus on ranking and recommendation. Property prediction was not thoroughly explored in these works; for example in (Qu et al., 2023), the text embeddings were frozen and used only for regression when it comes to property prediction. Another work that is closely related to ours was proposed by Korolev and Protsenko (2023). Similarly to Qu et al. (2023), they finetuned MatBERT on text descriptions of crystals, although the